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Raúl Mariani Galarza **Portfolio: raulmariani2000.github.io**

Mechanical and aerospace engineer, analog astronaut, including a space policy and entrepreneurship background pursuing human spaceflight engineering opportunities. U.S. citizen eligible for clearance.

EDUCATION

MSc in Space Studies - International Space University - Strasbourg, France Aug 2025 - Aug 2026

- Engineering & Applications, Space Policy & Entrepreneurship, Science and Human Exploration
- **Individual Project-** Viability of Spaceports for Latin America and Africa
- **Team Project-** Global Communication Roadmap for Deep Space Human Exploration

MEng in Aerospace Engineering - University of Cincinnati - Cincinnati, OH Aug 2024 - May 2025

- LH2 Power Systems, Intelligent Robotics, Fundamentals & Applications of Solar Energy
- **GPA:** 3.933

Double Major BS in Mechanical and Aerospace Engineering - Case Western Reserve University - Cleveland, OH Aug 2019 - May 2024

- Design and Manufacturing, Control of Mechanical Systems, Design of Mechanical Elements

WORK EXPERIENCE

MMAARS Trainer - Puerto Rico Space Foundation (PRSF) & Mars-Moon Astronautics Academy & Research Science (MMAARS) May 2025 - Present

- Taught, translated and graded Virtual Analog Astronautics Coursework to 84 graduates supporting successful completion of the MAAARS program in partnership with PRSF.
- Presented lectures on, Becoming an Analog Astronaut, MMAARS Astronautics, Extra Vehicular Activities, 3D printing in Space, AI training and data organization protocols and more.
- Performed research presented at the International Astronautical Congress 2026. The Resource Psychology Management (RPM) Project: Quantifying Human Resource Prioritization and Decision-Making Under Scarcity in Isolated and Extreme Analog Environments.

Vice Chair & Member at Large of SEDSUSA and Founder, President & Treasurer of SEDSCWRU - Students for the Exploration and Development of Space (SEDS) Mar 2020 - Nov 2024

- **Vice Chair & Member at Large of SEDSUSA:**
  - Led monthly organizational updates to the Board of Advisors.
  - Organized two SpaceVision conferences with over 650 attendees and over 100 speakers.
  - Sole Young Professional Representative through SEDSUSA at the Alliance for Space Development discussing national policy issues like preventing NASA budget reduction.
- **Founder, President & Treasurer of SEDSCWRU:**
  - Founded SEDS chapter at CWRU, complying with National and University Requirements.
  - Researched university funding regulations and prepared a successful funding request, securing the maximum $4,000 allocation per club to support attendance at SpaceVision.
  - Organized a total solar eclipse event at CWRU with activities for over 1,000 people

Co-op Engineer - Heroux Devtek - Strongsville, OH Aug 2022 - Aug 2023

- Led an R&D project to double plating capacity from two to four parts per cycle by designing, assembling, and drafting a custom tooling solution in CATIA. Collaborated directly with the manufacturer on design improvements and inspected the finished product after fabrication.
- Supported the engineering team with their different projects focused on increasing efficiency and decreasing points of failure found during inspection of the plating process.

Engineering Intern - WS Tyler - Mentor, OH May 2022 - Aug 2022

- Created digital parts, drawings and assembled the RO-TAP Sieve Shaker consisting of over 100 parts in Fusion 360 from physical parts and old drawings for preservation, backup and upgrades.

Deputy Project Manager - NASA L'Space Workforce Development Program Aug 2021–Dec 2021

- Served as Deputy Project Manager for an eight-member NASA L'SPACE team developing a Mars rover/drone mission concept focused on water resources, astrobiology, and human exploration. Supported engineering development of components, assemblies, and technical drawings.

CORE SKILLS

Tech: Catia, Fusion, SolidWorks, STK, MiniRefprop, MS Excel/PowerPoint/Word, Multisim, NX, JMars
Personal: English and Spanish, leadership, teamwork, budget tracking, data analysis